

BAUST - Department of Computer Science & Engineering

CSE 2105: Object Oriented Programming | Mid-Term Examination - Spring 2024 | Batch 12

Time: 1.5 Hours | Full Marks: 30

Question 1: OOP Principles & Polymorphism [10 Marks]

- a) Explain compile-time vs run-time polymorphism with code snippets in C++.
- b) Why are virtual destructors necessary when deleting derived objects through base pointers?

Question 2: Operator Overloading & Memory Allocation [10 Marks]

- a) Implement class 'Matrix2D' overloading operator+ and operator<< for stream output.
- b) Explain the Rule of Three / Rule of Five in modern C++ resource management.

Question 3: Exception Handling & Templates [10 Marks]

- a) Write a generic 'SafeStack<T>' template class that throws 'std::out_of_range' on pop when empty.
- b) Analyze the overhead of dynamic_cast vs static_cast in deep inheritance hierarchies.

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